

BUSINESS UNDERSTANDING REPORT.

Telco Customer Churn — Week 2

INTRODUCTION:

Customer churn is a significant concern for telecommunications companies because losing existing customers can affect revenue and business growth. Understanding customer characteristics and service usage patterns can help a company identify factors associated with customers leaving its services.

This project focuses on the Telco Customer Churn dataset, which contains customer demographic, service, contract, and billing information. The purpose of this Week 2 project is to understand the dataset and prepare it for future machine-learning development.

BUSINESS PROBLEM

The business problem addressed in this project is customer churn.

The telecommunications company needs to understand the characteristics and patterns of customers who leave its services. Before a reliable machine-learning model can be developed, the available customer data must be inspected, cleaned, transformed, and prepared appropriately.

Therefore, the focus of this stage is to ensure that the dataset is accurate, consistent, relevant, and suitable for subsequent machine-learning analysis.

BUSINESS OBJECTIVE

The primary objective of this project is to prepare the Telco Customer Churn dataset for future machine-learning modeling by identifying and addressing data-quality and preprocessing issues.

The specific objectives are to:

- Inspect the structure and quality of the dataset.
- Identify and handle missing values.
- Check and remove duplicate records where necessary.
- Examine data types and ensure they are appropriate.
- Perform relevant feature engineering.
- Analyze the distributions and relationships within the data through visualization.
- Detect potential outliers.

- Identify and remove irrelevant or highly redundant features.
- Convert categorical variables into numerical representations.
- Scale appropriate numerical features.
- Produce a clean and machine-learning-ready dataset.

DATASET AND BUSINESS RELEVANCE

The dataset contains 7,043 customer records and 21 original variables.

The variables provide information about:

Customer demographics

Customer tenure

Telephone services

Internet services

Additional services

Contract type

Billing methods

Monthly charges

Total charges

Churn status

The target variable is Churn, which indicates whether a customer has left the company's service.

These variables are relevant because customer characteristics, service subscriptions, contract arrangements, and billing information can provide useful information for understanding customer behavior.

KEY BUSINESS QUESTIONS.

The preprocessing and exploratory analysis are intended to help answer questions such as:

- What does the customer dataset contain?
- Are there missing or inconsistent values that could affect analysis?
- Are there duplicate customer records?

- Which features are relevant to understanding customer churn?
- Are there redundant or highly correlated features?
- Are there potential outliers in the numerical variables?
- What patterns can be observed from the distributions of customer and service variables?
- How can the categorical and numerical variables be transformed into a format suitable for machine learning?

EXPECTED BUSINESS VALUE

Properly preprocessing the dataset provides several potential benefits:

- ★ Improves the quality and reliability of the data.
- ★ Reduces errors that could affect future model performance.
- ★ Removes irrelevant or redundant information.
- ★ Makes categorical information usable by machine-learning algorithms.
- ★ Places numerical variables on appropriate scales.
- ★ Provides a consistent foundation for future churn prediction.

Ultimately, the prepared dataset can support the later development of a model that may help the company understand and predict customer churn.

PROJECT SCOPE.

This Week 2 project covers:

Data Inspection → Data Cleaning → Feature Engineering → Data Visualization → Outlier Detection → Feature Selection → Encoding → Scaling → ML-ready Dataset

The assignment specifically requires these preprocessing activities and the creation of a clean dataset ready for machine-learning use.

Model training, model comparison, hyperparameter tuning, and deployment are not part of this preprocessing stage.

SUCCESS CRITERIA.

The project will be considered successful when:

1. The dataset has been thoroughly inspected.

2. Missing values have been appropriately handled.
3. Duplicate records have been checked.
4. Data types are appropriate.
5. Relevant features have been engineered where necessary.
6. At least the required visualizations have been produced and interpreted.
7. Potential outliers have been investigated.
8. Irrelevant and highly redundant features have been addressed.
9. Categorical variables have been encoded.
10. Appropriate numerical variables have been scaled.
11. The final dataset contains no unresolved data-quality problems.
12. A clean machine-learning-ready dataset has been produced.

CONCLUSION.

The Week 2 project focuses on transforming the Telco Customer Churn dataset from its original form into a reliable dataset suitable for future machine-learning development.

Through systematic inspection, cleaning, visualization, feature engineering, outlier detection, feature selection, encoding, and scaling, the project addresses the major data-preparation requirements necessary before model development.

The final outcome of this stage is a clean, structured, numerical, and machine-learning-ready dataset, while actual model development is reserved for a later stage.